MYSQL ASSIGNMENT 1

DDL Commands and Constraints

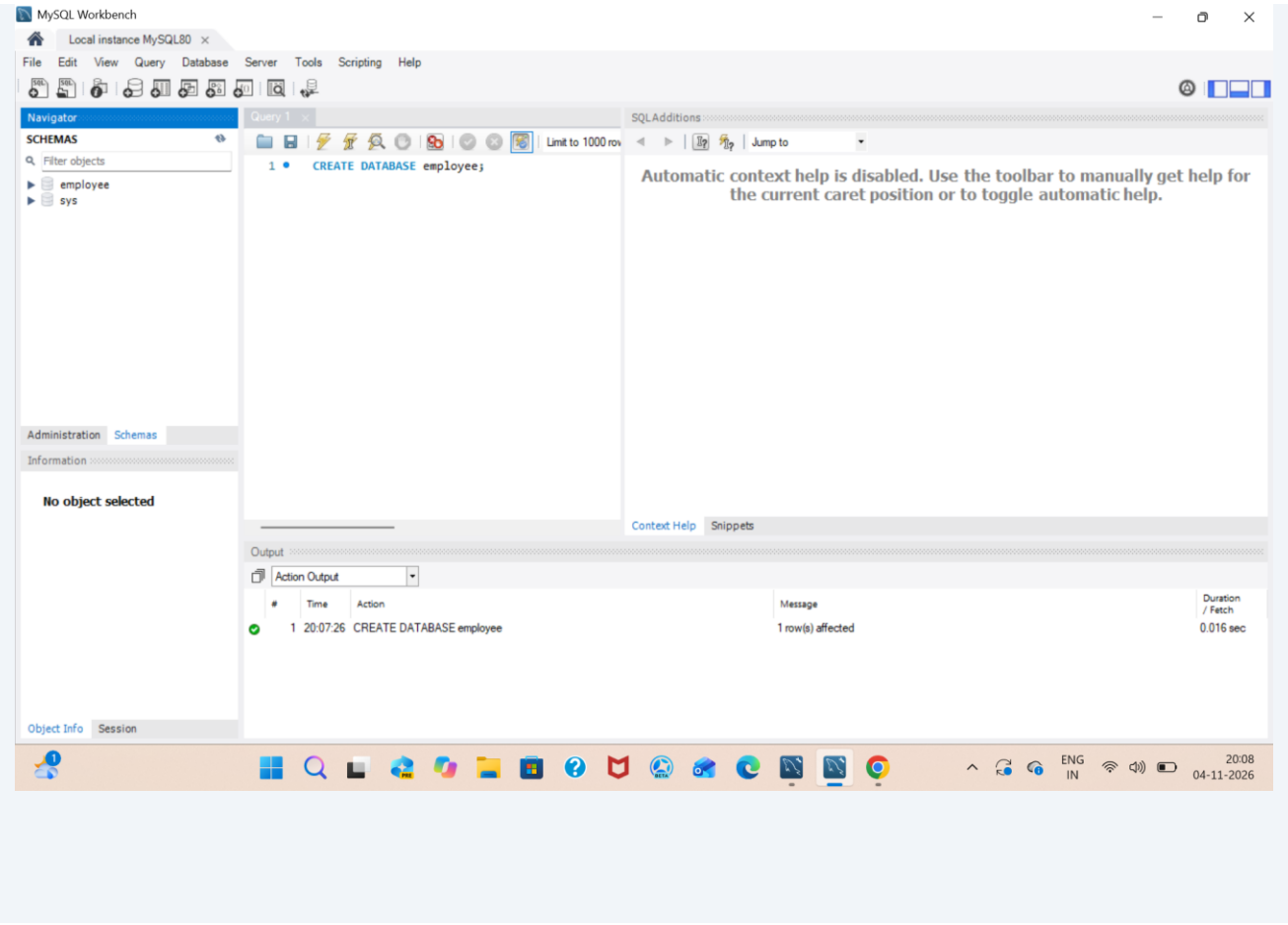
Introduction

This practical assignment was completed using MySQL Workbench. The work involved creating an employee database, changing table structures, renaming and removing database objects, and recreating the database with constraints to help maintain accurate and consistent data.

PART 1 – DDL COMMANDS

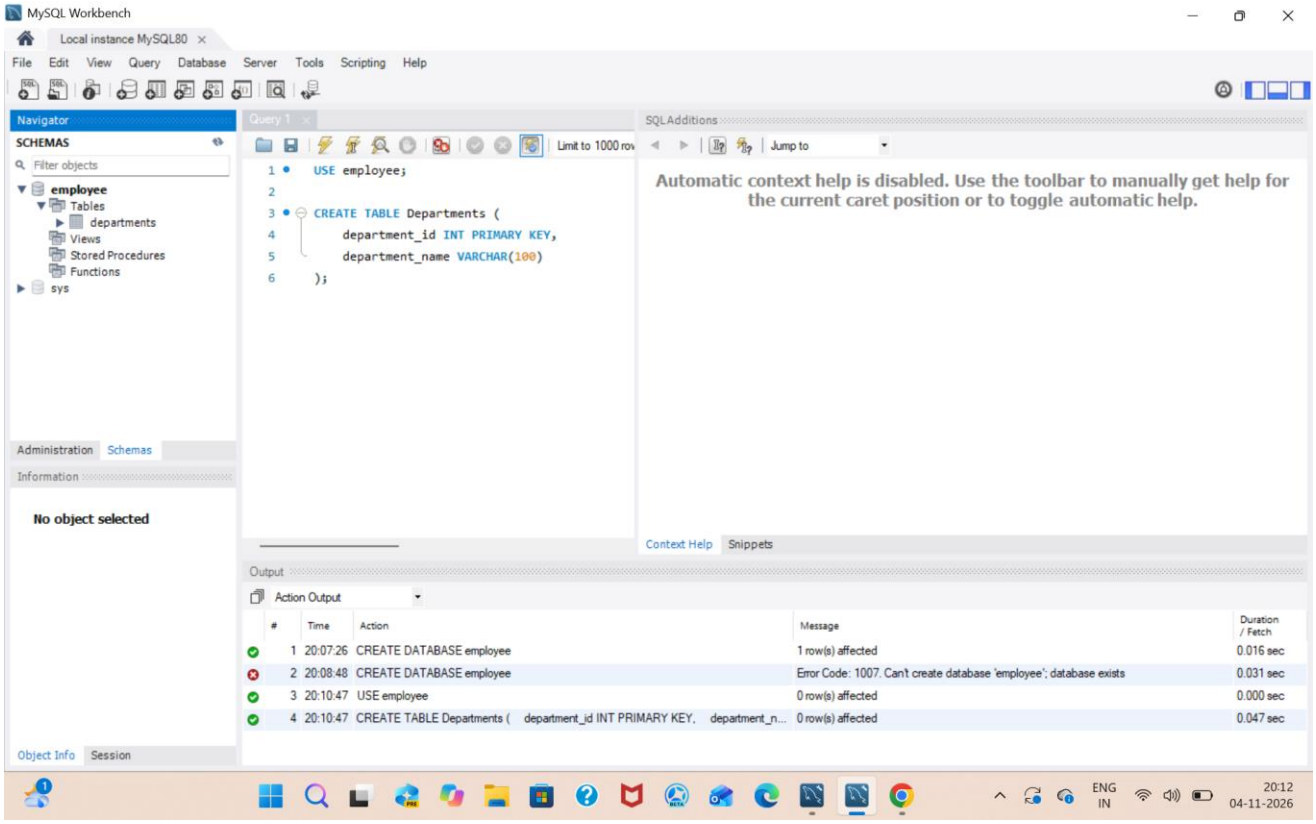
1. Creating the employee database and tables

A database named employee was created. Three tables were created inside it: Departments for department details, Location for location details, and Employees for employee information such as name, gender, age, hire date, designation, department, location and salary.



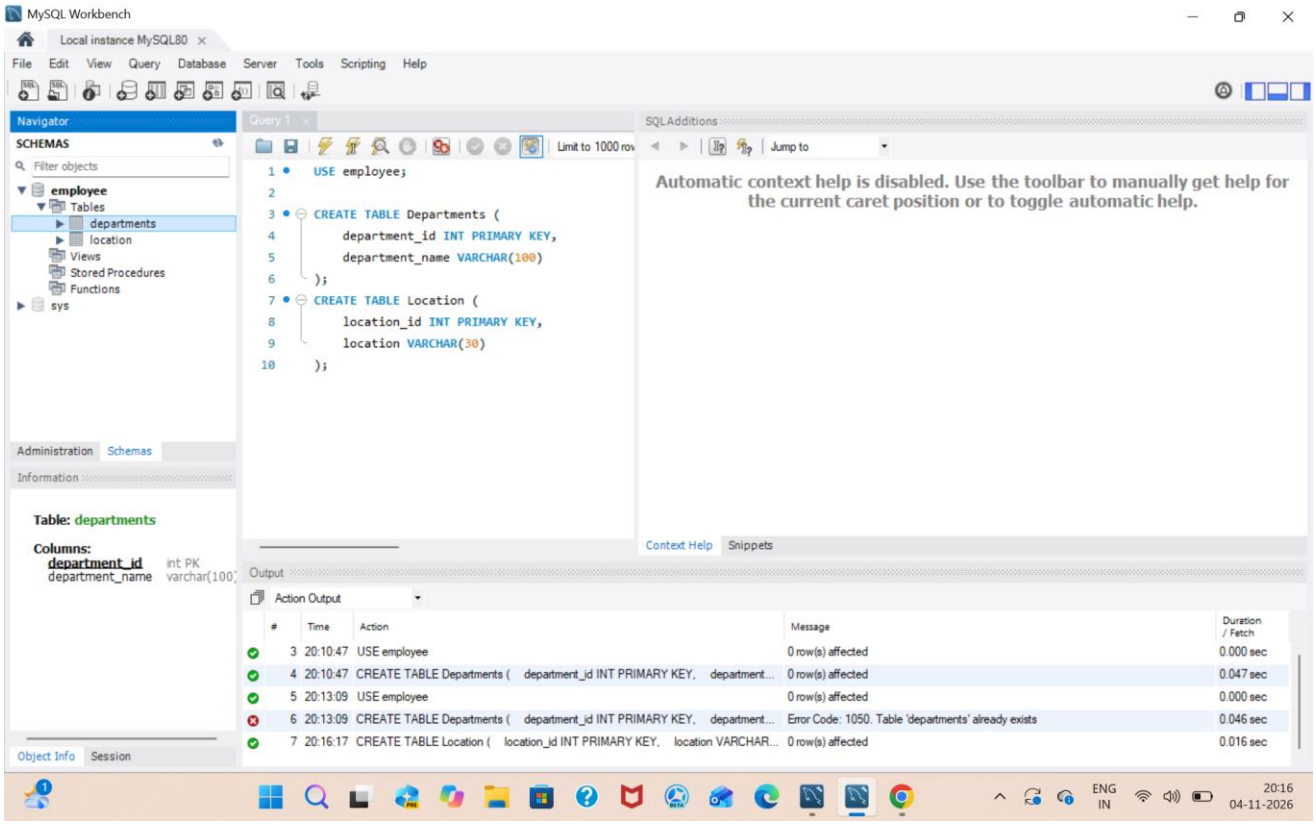
2. Altering the Employees table

The Employees table structure was changed as required. An email field was added for employee email addresses. The designation field was widened to store longer values. The age field was removed, and hire_date was renamed to date_of_joining.



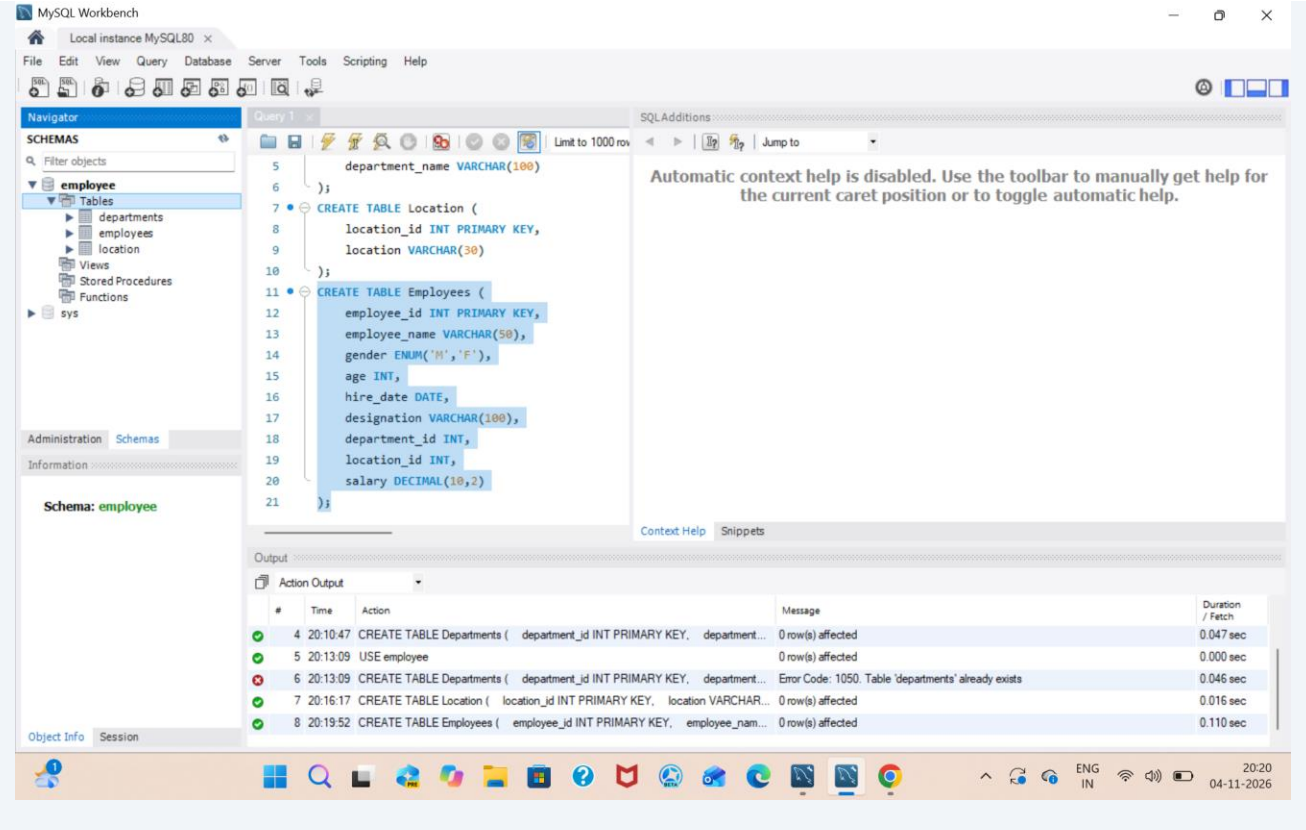
3. Renaming tables

The Departments table was renamed to Departments_Info, and the Location table was renamed to Locations. This demonstrated how existing tables can be renamed.



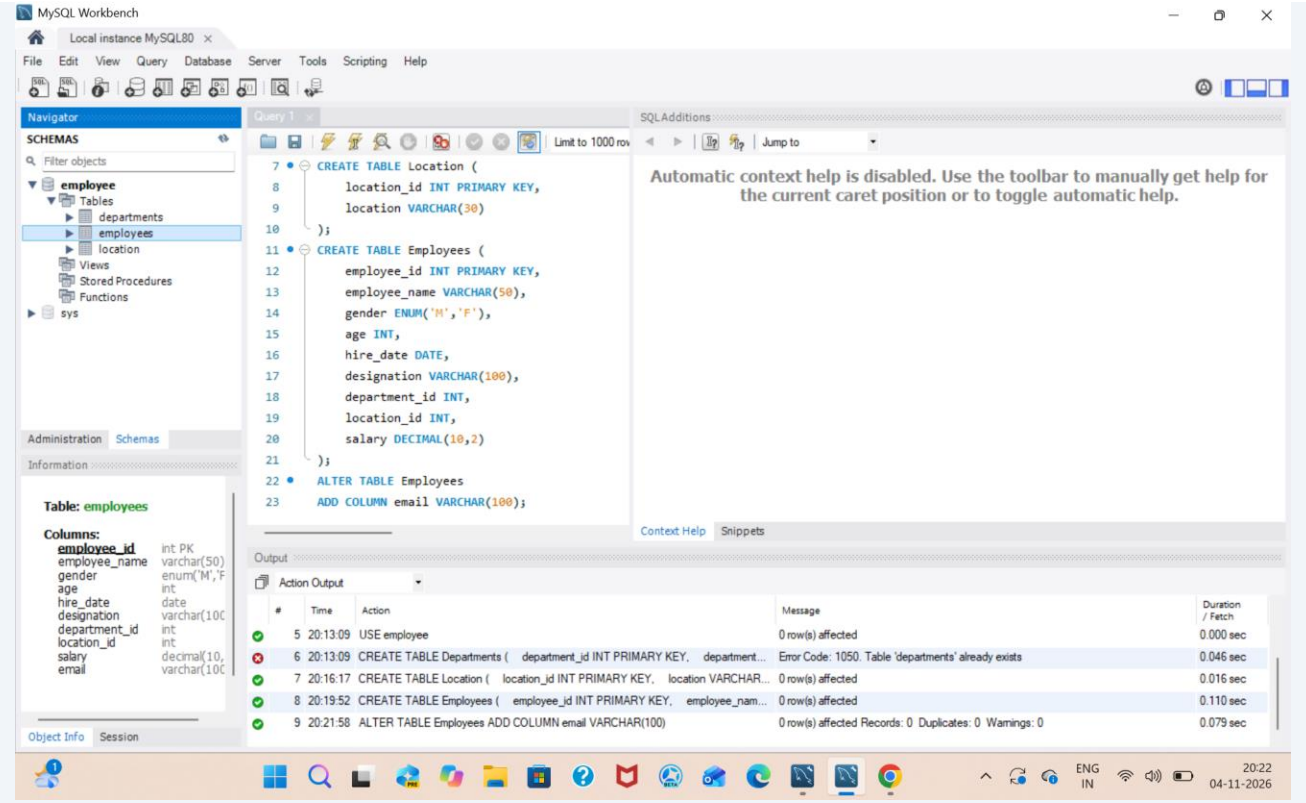
4. Truncating the Employees table

The Employee's table was truncated. This removed the records from the table while keeping its table structure.



5. Dropping the Employees table and database

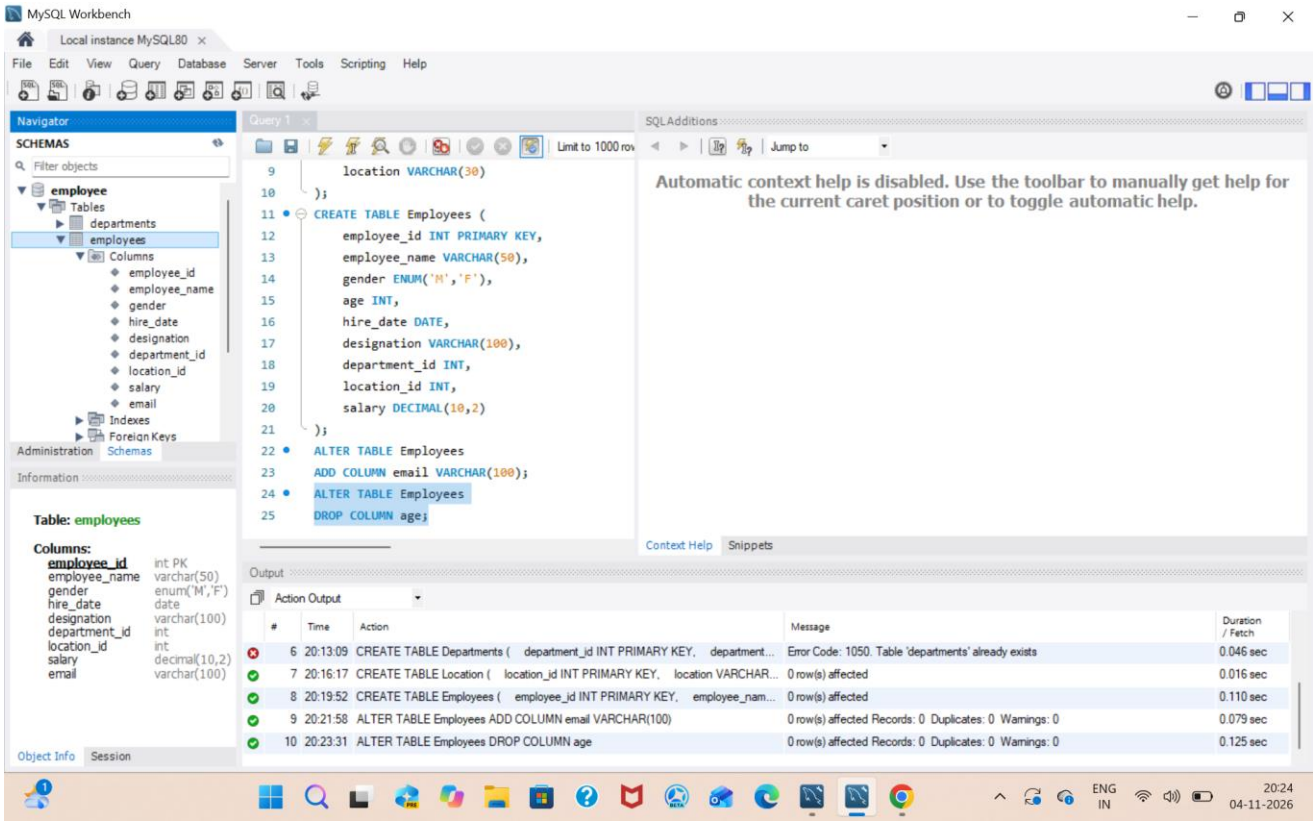
The Employee's table was dropped first. After that, the employee database was dropped, completing the removal operations in Part 1.


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
employee
Tables
departments
employees
location
Views
Stored Procedures
Functions
sys
Administration Schemas
Information
Table: employees
Columns:
employee_id int PK
employee_name varchar(50)
gender enum('M','F
age int
hire_date date
designation varchar(10
department_id int
location_id int
salary decimal(10,
email varchar(10
Object Info Session
Query 1
Limit to 1000 rov
7 CREATE TABLE Location (
8 location_id INT PRIMARY KEY,
9 location VARCHAR(30)
10);
11 CREATE TABLE Employees (
12 employee_id INT PRIMARY KEY,
13 employee_name VARCHAR(50),
14 gender ENUM('M','F'),
15 age INT,
16 hire_date DATE,
17 designation VARCHAR(100),
18 department_id INT,
19 location_id INT,
20 salary DECIMAL(10,2)
21);
22 ALTER TABLE Employees
23 ADD COLUMN email VARCHAR(100);
SQL Additions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Output
Action Output
Time Action Message Duration / Fetch
5 20:13:09 USE employee 0 row(s) affected 0.000 sec
6 20:13:09 CREATE TABLE Departments (department_id INT PRIMARY KEY, department... Error Code: 1050. Table 'departments' already exists 0.046 sec
7 20:16:17 CREATE TABLE Location (location_id INT PRIMARY KEY, location VARCHAR... 0 row(s) affected 0.016 sec
8 20:19:52 CREATE TABLE Employees (employee_id INT PRIMARY KEY, employee_nam... 0 row(s) affected 0.110 sec
9 20:21:58 ALTER TABLE Employees ADD COLUMN email VARCHAR(100) 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.079 sec
ENG IN
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PART 2 – CONSTRAINTS

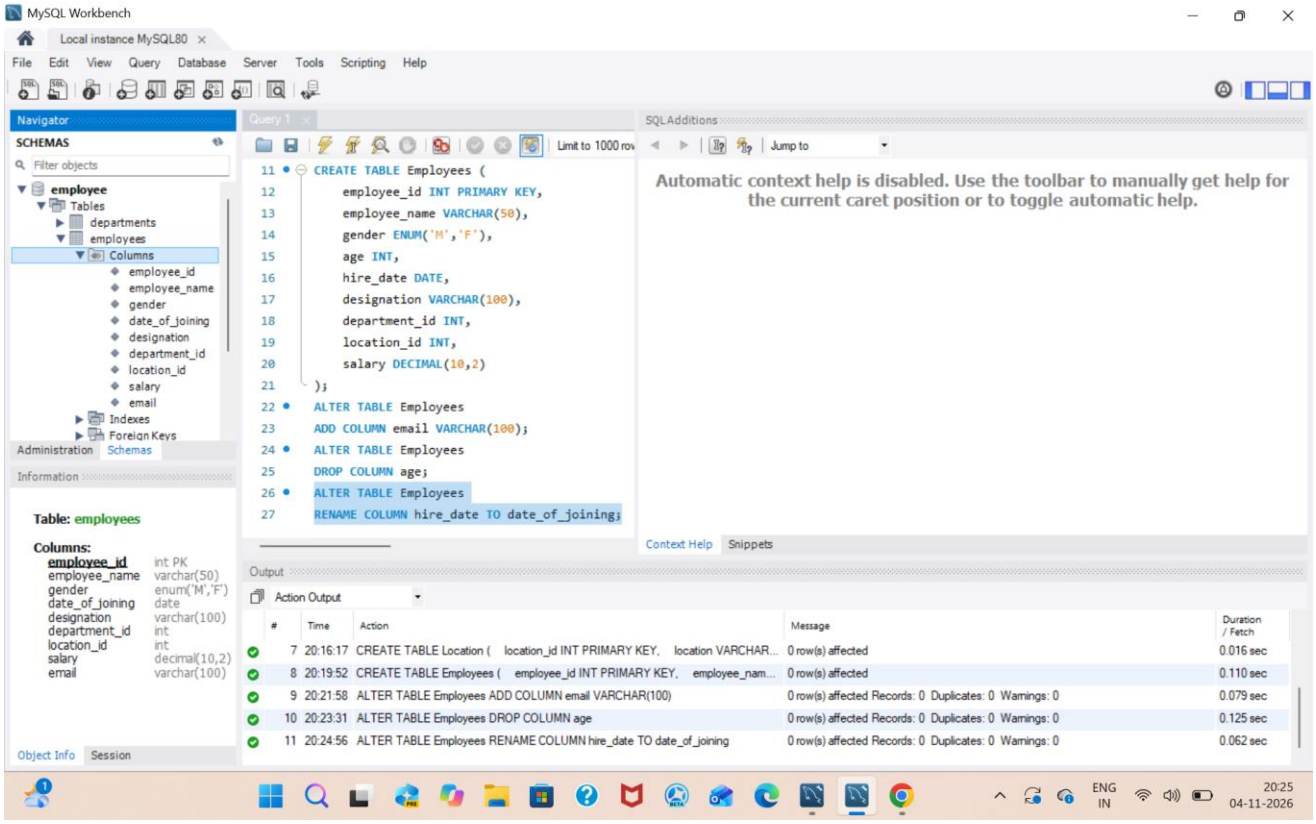
1. Recreating the employee database

After completing Part 1, the employee database was created again for the constraints section. The tables were recreated with the required structure and rules.



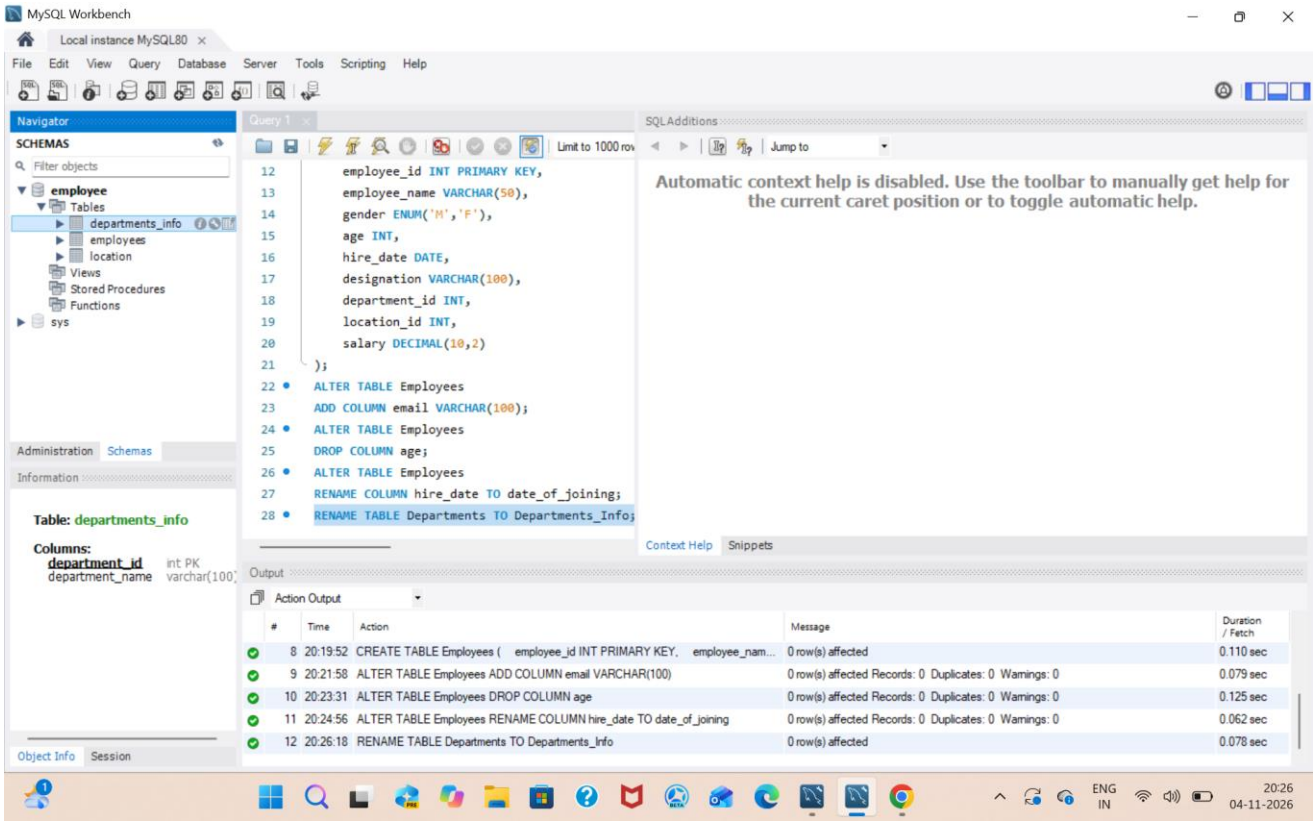
2. Departments table constraints

The Departments table uses department_id as its primary key, giving each department a distinct identifier. The department_name field is mandatory and unique, preventing missing or repeated department names.



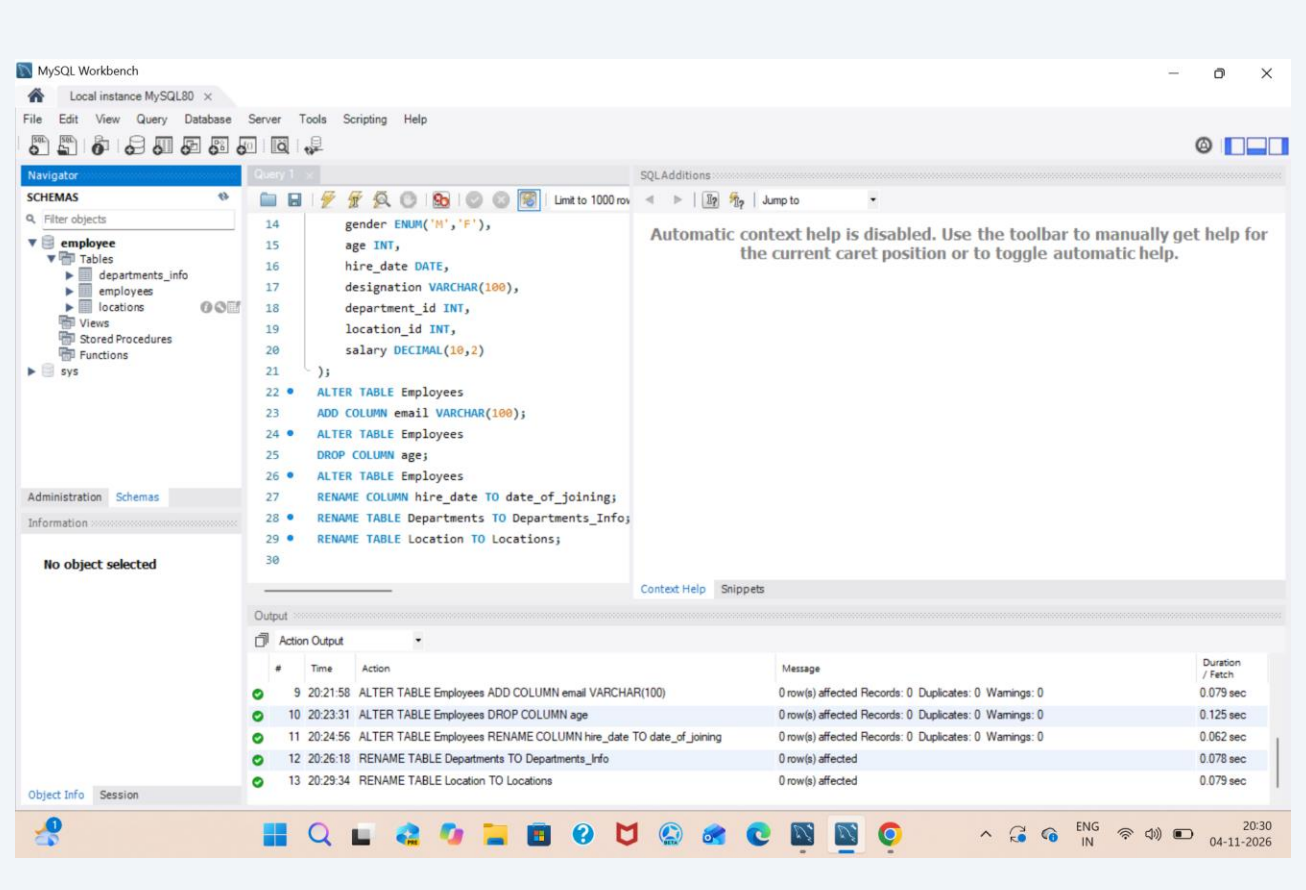
3. Location table constraints

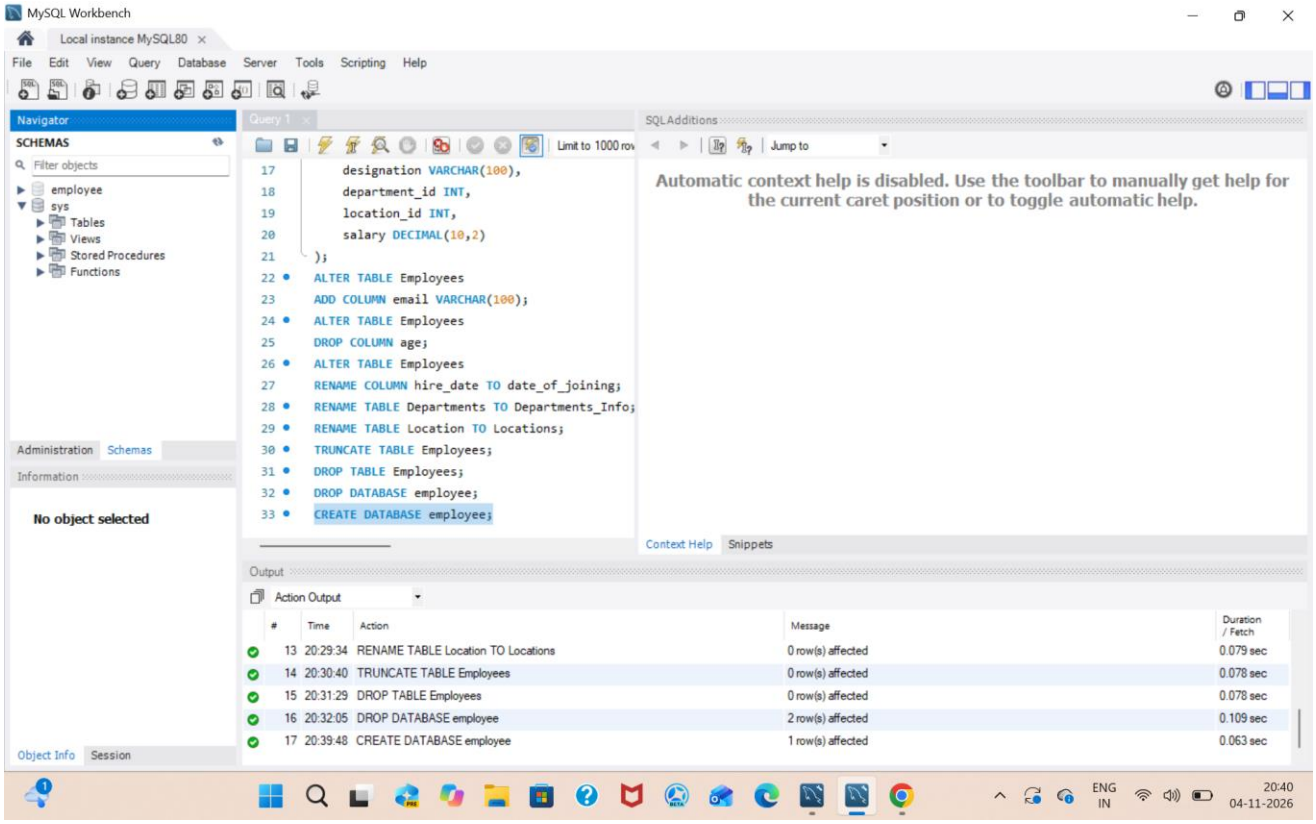
The Location table generates location_id values automatically in sequence. The identifier is a primary key. The location field is mandatory and unique, preventing blank or duplicate location names.



4. Employees table constraints

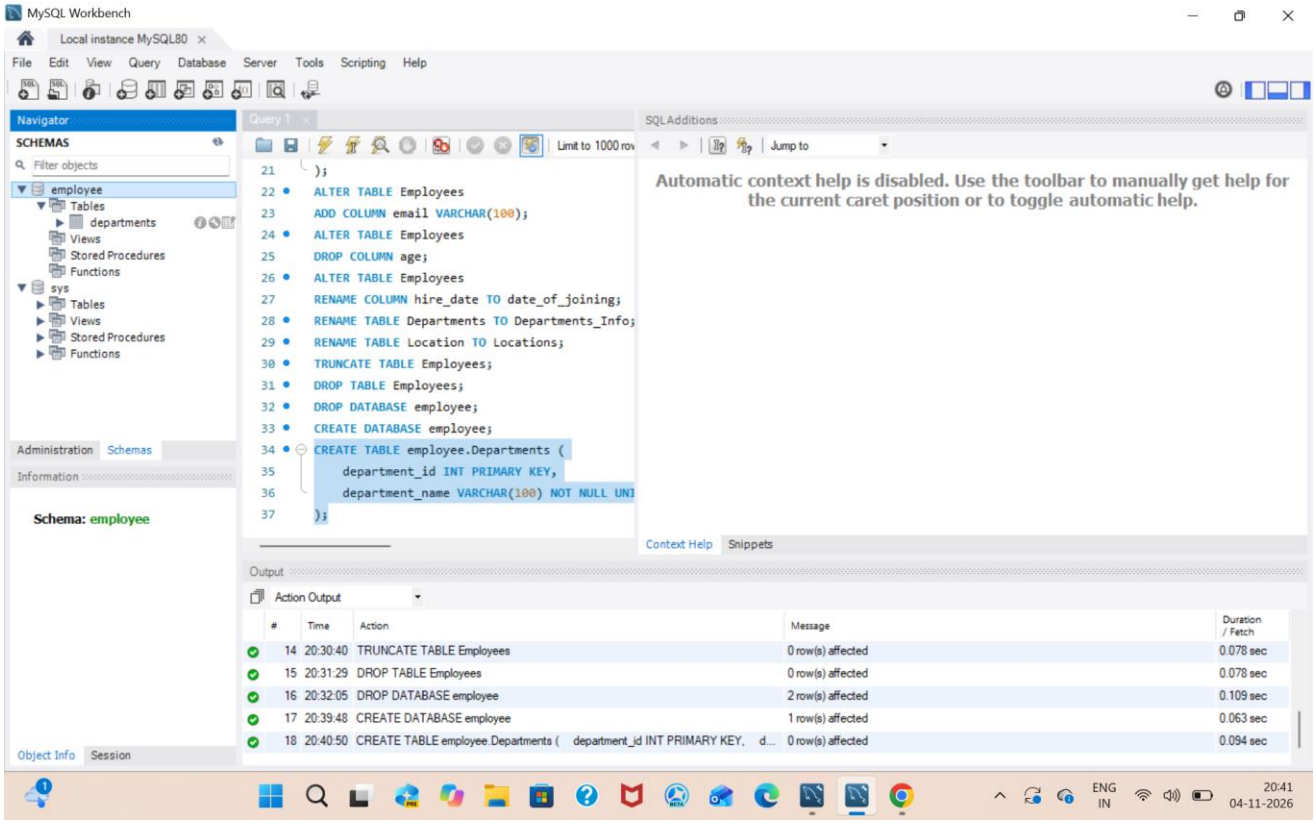
The Employees table gives each employee a distinct employee_id. The employee_name field is mandatory. The gender field is limited to M or F. A check condition requires age to be 18 or above. If hire_date is not supplied, the current date is assigned automatically. The department_id and location_id fields are linked to their respective tables using foreign-key relationships.





5. Final database structure

At the end of the work, the employee database contained the Departments, Location and Employees tables. The Employee's table was linked to the other tables through department and location references.



Conclusion

Through this assignment, I practised creating and managing a database and its tables in MySQL Workbench. I changed table structures, renamed tables, truncated a table, and dropped database objects. I then recreated the database and applied constraints to identify records, prevent missing or duplicate values, restrict allowed values, check age requirements, set a default hire date, and connect related tables. The screenshots in this report provide evidence of the practical work completed.